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Lab 12 Ethics, Fairness, and Bias in ML

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Reflection on Ethics, Fairness, and Bias in Machine Learning

In this lab, our group explored how machine learning models can produce different results for different groups of people. We used the Adult Income dataset and trained a Logistic Regression model to predict whether someone earns more than $50,000 a year. Unlike our previous labs, where we mainly focused on improving model performance, this assignment encouraged us to think about how those predictions could affect people in real situations. We learned that even a model with good accuracy can make unfair decisions if it produces different types of errors for certain groups.

In Part 2, Task 1, we calculated the model's accuracy for males and females separately to see how well it performed for each group (Tien). The overall accuracy was 84.61%, which initially seemed like a reasonable result (Tien). However, when we examined each group, we found that the accuracy was 81.20% for males and 91.81% for females (Tien). The difference of 10.61 percentage points showed us why looking only at overall accuracy can be misleading (Tien). At first, we might assume that higher accuracy means the model is fairer toward females, but we realized that accuracy alone cannot explain how the model treats each group (Tien). This made us question what other metrics we should consider before deciding whether a model is fair (Tien).

In Task 2, we used a confusion matrix to calculate the False Positive Rate (FPR) and False Negative Rate (FNR) for both groups (Viktoriya). The results showed that males had a higher FPR of 10.26%, compared to only 2.81% for females (Viktoriya). This means the model incorrectly predicted high income for low-income males more frequently (Viktoriya). On the other hand, females had a higher FNR of 47.84%, compared to 37.80% for males (Viktoriya). We found this comparison interesting because females had better overall accuracy but were more likely to receive incorrect low-income predictions when their actual income was high (Viktoriya). These results showed us that accuracy alone can hide important differences in how a model makes mistakes (Viktoriya). We connected these results to loan approvals and considered how incorrect predictions might create unfair advantages or disadvantages for different people (Viktoriya).

The hiring scenario in the Knowledge Check encouraged us to think more carefully about the consequences of prediction errors (Zaid). We discussed how a False Negative could cause an employer to reject someone who was actually qualified for a job (Zaid). Since the model had a higher FNR for females, we considered how this difference could lead to unfair disadvantages if

someone used similar predictions in hiring decisions (Zaid). This made us realize how important it is to carefully evaluate a model before using it to make decisions that affect people's opportunities (Zaid). Another concern was that the model predicted income rather than actual job qualifications, which made us question whether it would even be appropriate for hiring (Zaid).

Next, we explored different ways to reduce bias in the model (Zaid). One question asked whether removing the sex column would completely solve the problem (Zaid). Initially, this might seem like a simple solution because the model would no longer have direct access to gender information (Zaid). However, we learned that other features, such as occupation, education, and marital status, could still contain patterns related to gender (Zaid). This introduced the idea of proxy bias and helped us understand why removing a sensitive feature does not automatically make a model fair (Viktoriya). We discussed possible solutions, such as reweighting the data and adding fairness constraints, but we realized that we would still need to test the model again to check whether these changes actually reduced bias (Zaid).

This lab connected to our previous work with Model Evaluation and Data Preparation (Tien, Viktoriya, and Zaid). In earlier assignments, we used metrics such as accuracy, precision, recall, and confusion matrices to understand how well models made predictions (Viktoriya). This time, we used similar methods to compare how the model performed for different groups instead of only looking at its overall results (Tien). We now understand that evaluating a model requires more than checking whether its predictions are correct (Tien, Viktoriya, and Zaid). We need to consider which mistakes it makes, who might experience them, and what consequences they could have (Zaid). One question that remained was how we could balance overall accuracy with fairness if improving one metric caused another to become worse (Viktoriya).

In conclusion, this lab helped our group develop a better understanding of the ethical responsibilities involved in machine learning. By comparing accuracy and error rates for different groups, we saw how a model could have good overall results but still make more mistakes for certain people. The real-world scenarios made these differences easier to understand because they connected numerical results to decisions that could affect people's lives. We now recognize that fairness should be considered throughout the machine learning workflow, not just after training a model. In future projects, we want to pay more attention to subgroup performance and possible sources of bias before deciding whether a model is reliable enough for real-world use.